

Aurèle TOUSSAINT

CNRS Researcher (Chargé de Recherche, CR)

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PERSONAL INFORMATION

Date of birth:

16 July 1988

Nationality:

French

PROFESSIONAL EXPERIENCE

CNRS Researcher (CR) — CRBE, Université de Toulouse, France

2024 – present

Macroecology, functional diversity, freshwater fishes, vertebrates, biodiversity loss.

Research Fellow — Institute of Ecology and Earth Sciences, University of Tartu, Estonia

2020 – 2023

Estonian Research Council grant PSG-505 (Rank 1/42 · 260,500 EUR).

Topic: Toward an optimization of functional diversity approaches of freshwater fish: historical patterns and impact of human activities.

MOBILITAS Pluss Post-Doctoral Researcher — University of Tartu, Estonia

2017 – 2019

Estonian Research Council grant MOBJD-276 (Rank 1/106 · 63,300 EUR).

Topic: Macroecology of functional diversity: comparison of taxonomic groups.

Specialist Fellow in Macroecology — University of Tartu, Estonia

2016 – 2017

Project IUT20-29: Dark diversity: taxonomic, phylogenetic, functional and genetic levels in dynamic plant communities.

Erasmus-Mundus Research Visit — Kyung Hee University, Seoul, Republic of Korea

Jan–Jun 2016

Supervisor: Prof. Young-Seuk Park (Kyung Hee University).

PhD Student in Fish Macro-ecology — EDB Laboratory, University Paul Sabatier, Toulouse, France

2012 – 2016

Topic: Functional diversity of freshwater fish fauna at the world scale: patterns, determinants and impacts of human activities.

Supervisors: Prof. Sébastien Brosse (UPS Toulouse) & Dr. Sébastien Villéger (University Montpellier 2).

EDUCATION

PhD in Ecology — University Paul Sabatier-Toulouse 3, France

2016

Functional diversity of freshwater fish fauna at the world scale.

MSc in Biostatistics and Ecological Systems Modelling — UPS Toulouse 3, France

2012

Second highest distinction (Rank: 5/19). Thesis: Determinants of change in taxonomic dissimilarity in freshwater fish fauna at the world scale.

MSc (1st year) in Ecology — UPS Toulouse 3, France

2011

With distinction (Rank: 43/172). Thesis: Revision of the taxonomy of the freshwater fish European fauna and implications for conservation and biogeography.

BSc in Biology of Organisms — University of Lorraine, Nancy, France

2010

GRANTS & FUNDING

INTRAIT — Intraspecific trait variability in freshwater fish communities, French Guiana 2025 – 2026

Funded by CEBA & TREMPLIN (Labex CEBA). Principal Investigator.

CFBBA Chair — Building an Intraspecific Trait Database for Amazonian Freshwater Fishes 2026

Laureate of the inaugural Franco-Brazilian Chair Network (CFBBA). Collaboration with Prof. Luciano Montag (UFPA, Belém, Brazil).

Research Fellow — Estonian Research Council (PSG-505)

2020 – 2023

260,500 EUR — Rank 1/42.

MOBILITAS Pluss Post-Doctoral Fellow — Estonian Research Council (MOBJD-276) 2017 – 2019

63,300 EUR — Rank 1/106.

Erasmus-Mundus Mobility Grant for PhD Students

2016

Research visit at Kyung Hee University, Seoul, Republic of Korea.

AWARDS & DISTINCTIONS

Estonian Research Award in Natural and Exact Sciences

2024

National science prize awarded by the Estonian Academy of Sciences for research on the functional diversity of plants and vertebrates at global scale (Science Advances 2021; Nature Communications 2021).

PUBLICATIONS

* Co-senior authorship. Names in bold indicate Toussaint A.

Peer-reviewed articles in international journals

28. Chen Z., Xu C., Brosse S., **Toussaint A.**, Xu J., Su G. Global Patterns and Determinants of Multiple Facets of Freshwater Fishes Beta Diversity. *Global Ecology and Biogeography* (in press).
27. Partel M., Tamme R., [...], **Toussaint A.**, [...], Zobel M. Global impoverishment of natural vegetation revealed by dark diversity. *Nature*, 641, 917–924 (2025).
26. Coulon N., Pilet S., Lizé A., Lacoue-Labarthe T., Sturbois A., **Toussaint A.**, Feunteun E., Carpentier A. Shark critical life stage vulnerability to monthly temperature variations under climate change. *Marine Environmental Research*, 198, 106531 (2024).
25. Tordoni E., Carmona C.P., **Toussaint A.**, Tamme R., Pärtel M. Global patterns and determinants of multiple facets of plant diversity. *Global Ecology and Biogeography*, e13823 (2024).
24. **Toussaint A.**, Pärtel M., Carmona C.P. Contrasting impacts of non-native and threatened species on morphological, life-history, and phylogenetic diversity in bird assemblages. *Ecology Letters*, 27, e14373 (2024).
23. Coulon N., Lindegren M., Goberville E., **Toussaint A.**, Receveur A., Auber A. Threatened fish species in the Northeast Atlantic are functionally rare. *Global Ecology and Biogeography*, 32, 1128–1143 (2023).
22. Bueno C.G., **Toussaint A.**, Träger S. et al. Reply to: The importance of trait selection in ecology. *Nature*, 618, E31–E34 (2023).
21. de Tombeur F., Raven J.A., **Toussaint A.**, Lambers H., Cooke J., Hartley S.E., Johnson S.N., Coq S., Katz O., Schaller J., Violle C. Why do plants silicify? *Trends in Ecology and Evolution*, 38(3), 275–288 (2023).
20. Auber A., Waldock C., Maire A., Goberville E., Albouy C., Algar A.C., McLean M., Brind'Amour A., Green A.L., Tupper M., Vigliola L., Kaschner K., Kesner-Reyes K., Beger M., Tjiputra J., **Toussaint A.**, Violle C., Mouquet N.,

- Thuiller W., Mouillot D. A functional vulnerability framework for biodiversity conservation. *Nature Communications*, 13, 4774 (2022).
19. Paganeli B., **Toussaint A.**, Bueno C.G., Fujinuma J., Reier Ü., Pärtel M. Dark diversity at home describes the success of cross-continent tree invasions. *Diversity and Distributions*, 28(6), 1202–1213 (2022).
 18. Su G., Tedesco P.A., **Toussaint A.**, Villéger S., Brosse S. Contemporary environment and historical legacy explain functional diversity of freshwater fishes in the world rivers. *Global Ecology and Biogeography*, 31, 700–713 (2022).
 17. Brosse S., Charpin N., Su G., **Toussaint A.**, Herrera-R. G.A., Tedesco P.A., Villéger S. FISHMORPH: A global database on morphological traits of freshwater fishes. *Global Ecology and Biogeography*, 30(12), 2330–2336 (2021).
 16. Carmona C.P., Bueno C.G., **Toussaint A.**, Träger S., Diaz S., Moora M., Munson D.A., Pärtel M., Zobel M., Tamme R. Fine-root traits in the global spectrum of plant form and function. *Nature*, 597(7878), 683–687 (2021).
 15. Carmona C.P., Tamme R., Pärtel M., de Bello F., Brosse S., Capdevila P., Gonzalez-M R., Gonzalez-Suarez M., Salguero-Gomez R., Vasquez-Valderrama M., **Toussaint A.** Erosion of global functional diversity across the tree of life. *Science Advances*, 7(13), eabf2675 (2021).
 14. **Toussaint A.**, Brosse S., Bueno C.G., Pärtel M., Tamme R., Carmona C.P. Extinction of threatened vertebrates will lead to idiosyncratic changes in functional diversity across the world. *Nature Communications*, 12, 8182 (2021).
 13. Derezal O., Mondy P.C., Dembski S., Kreutzenberger K., Reyjol Y., Chanderis A., Valette L., Brosse S., **Toussaint A.**, Beillard J., Merg M.L., Usseglio-Polatera P. A diagnostic-based approach to assess specific risks of river degradation in a multiple pressure context: Insights from fish communities. *Science of the Total Environment*, 734, 139467 (2020).
 12. **Toussaint A.**, Bueno C.G., Davison J., Moora M., Tedersoo L., Zobel M., Öpik M., Pärtel M. Asymmetric assembly processes in plant and arbuscular mycorrhizal fungal communities across a grassland productivity gradient. *Journal of Vegetation Science*, 31(2), 355–366 (2020).
 11. **Toussaint A.**, Charpin N., Villéger S.* & Brosse S.* Worldwide freshwater fish homogenization is driven by few widespread non-native species. *Ecology Letters*, 21(11), 1649–1659 (2018).
 10. Kuczynski L., Côte J., **Toussaint A.**, Brosse S., Buisson L., Grenouillet G. Spatial mismatch in morphological, ecological and phylogenetic diversity in historical and contemporary European freshwater fish faunas. *Ecography*, 41, 1665–1674 (2018).
 9. Tedersoo L., Laanisto L., Rahimlou S., **Toussaint A.**, Hallikma T., Pärtel M. Global database of plants with root-symbiotic nitrogen fixation: Nod DB. *Journal of Vegetation Science*, 29(3), 560–568 (2018).
 8. **Toussaint A.**, Charpin N., Brosse S.* & Villéger S.* Global functional diversity of freshwater fish is concentrated in the Neotropics while functional vulnerability is widespread. *Scientific Reports*, 6, 22125 (2016).
 7. **Toussaint A.**, Beauchard O., Oberdorff T., Brosse S., Villéger S. Worldwide freshwater fish homogenization is driven by a few widespread non-native species. *Biological Invasions*, 18, 1295–1304 (2016).
 6. Allard L., **Toussaint A.**, Vigouroux R., Brosse S. Length-weight relationship of 58 fish species in streams of French Guiana. *Journal of Applied Ichthyology*, 31, 567–570 (2015).
 5. **Toussaint A.**, Beauchard O., Oberdorff T., Brosse S., Villéger S. Historical assemblage distinctiveness and the introduction of widespread non-native species explain worldwide changes in freshwater fish taxonomic dissimilarity. *Global Ecology and Biogeography*, 23, 574–584 (2014).

Preprints

4. Tordoni E., **Toussaint A.**, Pärtel M., Nogues-Bravo D., Carmona C.P. Combining taxonomic, phylogenetic and functional diversity reveals new global priority areas for tetrapod conservation. *bioRxiv* (2021). doi: 10.1101/2021.07.01.450689
3. **Toussaint A.**, Pärtel M., Carmona C.P. Non-native bird species will not compensate for the loss of phylogenetic and functional diversity after the extinction of threatened species. *Authorea* (preprint).

CONFERENCE PRESENTATIONS

Oral presentations

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| GfÖ Annual Meeting for Macroecology, Marburg, Germany | 2024 |
| Toussaint A., Pärtel M., Carmona C.P. Contrasting impacts of non-native and threatened species on morphological, life-history, and phylogenetic diversity in bird assemblages. | |
| 19th ISRLE, China | 2023 |
| Carmona C.P. et al. Mapping extinction risk in the global functional spectra across the tree of life. | |
| GfÖ Annual Meeting for Macroecology, Konstanz, Germany | 2020 |

Carmona C.P., Tamme R., Pärtel M., de Bello F., Brosse S., Capdevila P., Gonzalez-M R., Gonzalez-Suarez M., Salguero-Gomez R., Vasquez-Valderrama M., Toussaint A. Mapping extinction risk in the global functional spectra across the tree of life.

GfÖ Annual Meeting for Macroecology, Birmendorf, Switzerland 2018

Toussaint A., Pärtel M. Global mismatch between the species richness of vascular plants and symbiosis fungi across biogeographic realms.

60th IAVS Annual Symposium, Palermo, Italy 2017

Toussaint A., Pärtel M. Global mismatch between the species richness of vascular plants and symbiosis fungi across biogeographic realms.

7th EAFES International Congress, Daegu, Republic of Korea 2016

Toussaint A., Charpin N., Brosse S., Villéger S. Functional diversity and vulnerability of freshwater fish at the global scale.

9th SEFS, Geneva, Switzerland 2015

Toussaint A., Charpin N., Brosse S., Villéger S. Introductions of non-native freshwater fish affected differently the taxonomic and functional biodiversity facets.

8th SEFS, Münster, Germany 2013

Toussaint A., Beauchard O., Oberdorff T., Brosse S., Villéger S. Historical assemblage distinctiveness and the introduction of widespread non-native species explain worldwide changes in freshwater fish taxonomic dissimilarity.

Poster presentations

7th IBS Biennial Conference, Bayreuth, Germany 2015

Toussaint A., Charpin N., Brosse S., Villéger S. Functional diversity of the freshwater fish fauna.

TEACHING

Teaching assistant — EUR TULIP, University of Toulouse, France 2024 – present

- Functional ecology (MSc degree)

Invited lecturer — Universidade Federal do Pará (UFPA), Belém, Brazil 2026

- Introduction to functional ecology in R: trait data, functional diversity indices, RLQ & fourth-corner analyses, fish landmark morphometrics.

Jury member — Master Écologie Évolutive (EE), University of Toulouse 2025 – present

- Member of the examination jury for Master 2 students in Evolutionary Ecology.

Teaching assistant — University of Tartu, Estonia 2019 – 2023

- Data analysis in community ecology (BSc, MSc, and PhD students)

Teaching assistant — University Paul Sabatier – Toulouse 3, France 2013 – 2015

- Ecology & Quantitative Biology (BSc degree)
- Biostatistics (BSc degree)
- Multivariate analyses (MSc degree)

STUDENT SUPERVISION

PhD students

Pierre Bouchet 2024 – 2027

Functional diversity of vertebrates and human use: links between functional traits, anthropogenic exploitation and the biodiversity crisis.

Master students

Adrien Lagalle 2026

Comparison of methods for measuring functional traits of freshwater fishes.

Anouk Schneider 2026

Quantification of user bias in morphological trait measurement of freshwater fishes.

Pierre Bertrand 2026

Quantification of intraspecific diversity of freshwater fishes in French Guiana and its impact on functional diversity estimates.

Nell Pause 2025

Disentangling morphological, life-history, and dietary predictors of extinction risk and population trends in birds.

Adrien Solacroup 2014

Characterization of morphological intraspecific variability in French Guianese and European freshwater fish communities.

Bachelor students

Laura Boutic 2026

Quantification of user bias in morphological trait measurement of freshwater fishes.

Joseph Crochez 2026

Quantification of user bias in morphological trait measurement of freshwater fishes.

LANGUAGES

French	Mother tongue
English	Fluent (excellent command)
Italian	Basic communication

TECHNICAL SKILLS

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- Statistical analysis, programming and advanced graphics: R (expert level)
 - Large-scale biodiversity databases and trait databases
 - Multivariate statistics, functional diversity indices
 - GIS — ArcGIS
 - Image processing and landmark-based morphometrics
 - Office software (Windows / Linux / macOS)
 - Driving licence